



Towards improved automatic chemical kinetic model reduction regarding ignition delays and flame speeds

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ABSTRACT

In chemical kinetic model reduction under internal combustion engine conditions, most implementations only consider ignition related chemistry without consideration of flame speed prediction. In practice, flame propagation commonly exists in spark ignition engines, dual-fuel with pilot injection compression ignition engines, reactivity controlled compression ignition engines, and etc. Due to the inherent time-consuming nature, it is impractical to run a 1-D flame code with trial-and-error methods for model reduction, especially when starting with a large chemical kinetic model. In this paper, an improved reduction methodology is proposed for construction of a small set of species that give accurate predictions of both flame speeds and ignition delays. First, a strong correlation is found between the errors of maximum H radical and the errors in prediction of laminar flame speeds. Addition of H to the search targets in graph-based methods is conducted showing improvement in accuracy of flame speed prediction. Second, the normalized flame speed sensitivity with rate constants is analyzed for identifying a set of species that strongly influences the prediction of flame speeds. Finally, a trial-and-error based method is used for further reduction with a 0-D testbed for prediction of ignition only, while keeping the species important to flame chemistry. The newly proposed reduction methodology is used for development of accurate skeletal models predicting both ignition and flame speeds for several hydrocarbon fuels. These skeletal models include methane (27 species), propane (32 species), n-heptane (126 species), and primary reference fuel gasoline surrogates (207 species) with high fidelity to be used in engine simulations.

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1. Introduction

Chemical kinetic modeling plays an essential role in reactive flow simulations, which are routinely applied in assisting designs of efficient combustion devices. However, large-scale simulations with detailed reaction mechanisms are prohibitively expensive in terms of memory and CPU time. Stiffness caused by the wide range of timescales is another issue limiting the use of large reaction mechanisms. Therefore, it is imperative to reduce the complexity of the chemical kinetic model by retaining sufficient species and reaction steps to adequately reproduce relevant physical phenomena under target conditions [1]. Various approaches have been proposed and combined to determine the unimportant species or reaction pathways in a detailed reaction model [1,2]. As the demand to develop a fairly small and accurate reduced model for a particular simulation grows, how to effectively recognize redun-

dant species and reactions according to our needs is the goal to be achieved during the model reduction.

Compared to the approaches with hierarchical group combination by sensitivity analysis or graphical reaction pathway analysis [3–6], an automatic reduction technique is much more efficient and intelligent requiring little effort and expert knowledge from the user. With the detailed chemistry information under the targeted conditions of interest, a small set of species satisfying a pre-set level of error can be identified automatically without the need to tune the species list after running the software program. In the last decade, several improvements were made in the development of skeletal model by removing the redundant species and their associated reactions. The graph-based methods, such as the directed relation graph (DRG) [7], DRG with error propagation (DRGEP) [8], and path flux analysis (PFA) [9], are often employed as the first-step reduction due to their efficient features. After a satisfactory skeletal model has been developed, say using a DRG based method, further reduction is possible by using trial-and-error methods, such as Target Search Algorithm (TSA) [10], DRG-aided sensitivity analysis (DRGASA) [11], and etc. The main difference among various methods is how to rank the species list to decide which species

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is to be checked first. Niemeyer et al. [12] combined multiple methods into an automatic reduction scheme, the mechanism automatic reduction software (MARS). Lu et al. [13] extended the DRG method with expert knowledge (DRGX) by adding targets of species concentrations, allowing smaller reduction errors for the species of interest while larger errors for other species. Also, packages with other reduction schemes were developed such as simulation error minimization (SEM) [14]. In this study, we will mainly improve the reduction methodology combining graph-based methods and trial-and-error methods. Chen and Chen [15] demonstrate that for a fixed error, the combined scheme can achieve a large degree of reduction than simply using the trial-and-error method.

Under the condition of interest, the chemistry information of the detailed mechanism needs to be sampled for the graph-based method to calculate pairwise relation between species from a testbed. A closed homogeneous reactor [16], a perfectly stirred reactor [17], a premixed flame [18], or a Homogeneous Charge Compression Ignition (HCCI) simulation [19], can be used as the testbed in model reduction. In general, running a flame code is time-consuming with large detailed chemistry models to extract samples under a wide range of thermochemical states. The skeletal models generated by the graph-based approaches are often still large, thus further reduction conducted by the trial-and-error methods to assess whether the error induced by removing a species is unacceptable. The repeated running for all targeted conditions will make the reduction work prohibitively lengthy if one case takes too much time to run. The zero-dimension (0-D) closed homogeneous reactor is advantageous over others due to its efficient feature by evaluating global parameters such as ignition delay time.

Most past development of reduced chemical kinetic models under engine conditions doesn't take low-temperature ignition and flame propagation into account simultaneously [12,16,20,21]. Auto-ignition delay time under high pressure, especially in the negative temperature coefficient (NTC) regions [4,6,16], are the most important for HCCI engines. In contrast, flame propagation is the most salient feature in combustion devices of spark ignition engines. In new combustion concepts in engines, such as dual-fuel with pilot injection compression ignition engines, and reactivity controlled compression ignition engines, flame propagation can be also important under certain operating conditions. More importantly, most flamelet-based tabulation methods [22] require fast computations of laminar flames over the engine conditions. In this study, the existing issues in current automatic reduction approaches considering flame speeds are discussed first. Then, an improved automatic reduction methodology is proposed to accurately retain both flame speed and ignition related chemistry. To demonstrate the proposed improvement, several fuel kinetic models including methane, propane, n-heptane and primary reference fuel (PRF) gasoline surrogates are reduced and validated.

2. Automatic chemical kinetic model reduction methodology

2.1. Directed relation graph with error propagation (DRGEP) method

Due to their efficient feature, graph-based methods are now routinely used for development of skeletal models from detailed mechanisms. In this study, the DRGEP approach (Pepiot-Desjardins and Pitsch [8]) is used as a representative method. Several studies [12,15] have shown that the DRGEP method can achieve more accurate searching than the earliest graph-based method, namely, the DRG method developed by Lu and Law [7] but at the price of losing control over the worst-case errors. The detailed procedures in the DRGEP reduction are summarized as follows.

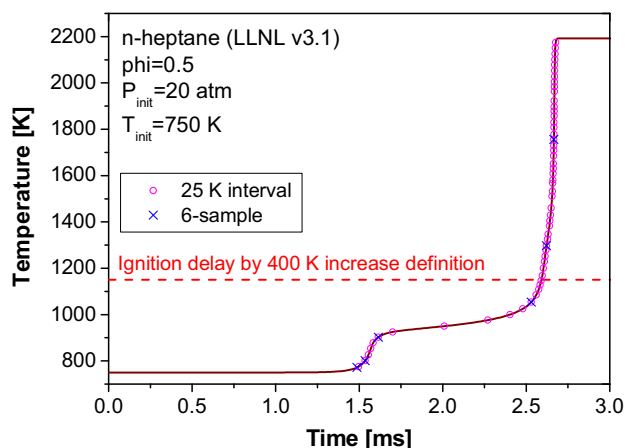


Fig. 1. An example case of 0-D n-heptane simulation using the WMR model with the scatters of x-mark for 6-sample extraction, and circles for 25 K temperature interval sampling. The detailed model is the LLNL v3.1 [24] with 654 species.

- (1) Chemical states of a certain combustion process need to be computed and sampled. For HCCI applications, the auto-ignition delays are most relevant features to be reproduced accurately by the reduced model. An in-house computer code for a well-mixed reactor (WMR) [23] is used for computing auto-ignition delays and writing input files for automatic reduction code. Herein, the issue of adequate sampling chemical states during auto-ignition simulations is discussed. Figure 1 shows a typical n-heptane auto-ignition case under the conditions of equivalence ratio (ϕ) of 0.5, initial temperature and pressure of 750 K and 20 atm, respectively. The Lawrence Livermore National Laboratory (LLNL) v3.1 n-heptane mechanism [24] with 654-species is used in the simulation. The auto-ignition delay time is defined as 400 K above the initial temperature. Six chemical states (denoted by x symbols) at temperatures of 20 K, 50 K, 150 K, 300 K, 550 K, and 1000 K increment above the initial temperature were chosen to cover low and high temperature combustion processes. The circles represent the uniform 25 K temperature interval sampling. In order to examine the difference between the two types of sampling methods, thirty target cases including the NTC regions are selected in the range of $P_0 = \{20 \text{ atm}, 40 \text{ atm}\}$, $\phi = \{0.5, 1.0, 2.0\}$ and $T_0 = \{750 \text{ K}, 825 \text{ K}, 875 \text{ K}, 950 \text{ K}, 1100 \text{ K}\}$ for the DRGEP reduction. Figure 2 plots the worst case relative error (by comparing the detailed and skeletal chemical model running among the thirty cases) against the species number of each skeletal model during the reduction. Based on the past experience and analysis on the reduction process [15,25], a 10% relative error tolerance which is a post-reduction criterion to select a desired skeletal model is considered reasonable for subsequent reduction. The plot indicates that small differences exist in the DRGEP reduction between the two sampling methods when the 10% error tolerance is considered; thus the 6-sample scheme is found adequate and will be used in each target case in this study.
- (2) The pairwise relation between species needs to be determined using the sampled chemistry information. In this study, the Jacobian defined pairwise relation shown in Eq. (1) [15] is used with n_{sp} denoting total number of species in the detailed model. The Jacobian matrix elements are formulated by defining a semi-normalized sensitivity coefficient, which is solved numerically using Eqs. (2)–(4). Y_i and Y_j are the mass fraction of the i th and j th species. W_i and ω_i are the molecular weight and molar production rate of the

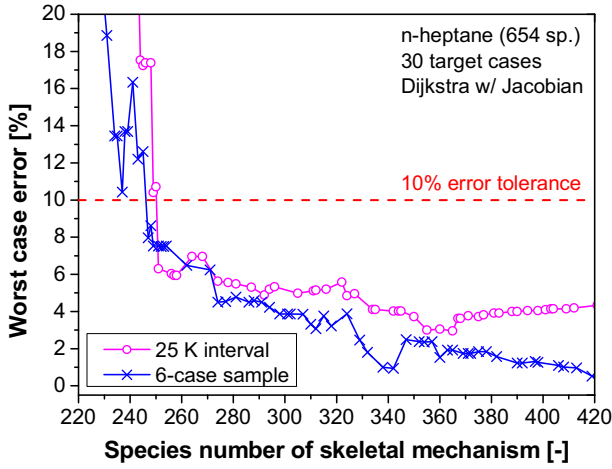


Fig. 2. Worst case error among 30 target cases against species number in each skeletal model during the DRGEP reduction process.

ith species, respectively, and ρ is the density. In comparison with other definition, the Jacobian-aided pairwise relation has been shown to improve the skeletal reduction performance [15,25,26]. By perturbation of the mass fraction of species B, the changed production rate of species A is contributed by all the reactions involving species B. In contrast, the pair-wise relations defined by others [7,8,27] only consider the local interactions with reactions including both species A and B.

$$r_{AB}^{JAC} = \frac{|J_{A,B}|}{\max_{1 \leq j \leq n_{sp}} (|J_{A,j}|, 10^{-40})} \quad (1)$$

where

$$J_{i,j} = Y_j \frac{\partial}{\partial Y_j} \frac{dY_i}{dt} \quad (2)$$

$$= Y_j \frac{W_i}{\rho} \frac{\partial \dot{\omega}_i}{\partial Y_j} \quad (3)$$

$$= Y_j \frac{W_i}{\rho} \frac{\dot{\omega}_i(P, T, Y_j + \Delta Y_j) - \dot{\omega}_i(P, T, Y_j)}{\Delta Y_j} \quad (4)$$

- (3) After determining the pairwise relations among species, a graph search algorithm is used to construct the reaction pathway starting from user-defined key species such as fuel, oxidizers, products, and intermediate radicals. Several studies have assessed the effects of graph search algorithms on the reduction outcomes [15,19,28] and the Dijkstra's algorithm was shown to be superior to others. Hence, the Dijkstra's algorithm is used for searching, and details can be found in [15].
- (4) Finally, a non-dimensional value (between 0 and 1) is assigned to each species based on the pairwise relation and the pathway for determining if a species is important. If the non-dimensional value of a species exceeds a cut-off threshold, the species will be kept in the skeletal model; otherwise the species will be removed along with its associated reactions. For each targeted condition, one can develop a skeletal model. The final skeletal model is the union of skeletal models for each target case. Although the relation between the relative error and the cut-off threshold is nonlinear, setting a higher threshold hold usually leads to a smaller skeletal model. Depending on the nature of the detailed chemical model and the desired accuracy, the threshold is set in the range from 10^{-5} to 0.1.

Table 1

Target conditions for compression ignition engine simulation in 0-D testbeds.

	ϕ	T [K]	P [atm]
$t_{ign}-1$	0.2	750	10
$t_{ign}-2$	0.2	750	80
$t_{ign}-3$	0.3	800	40
$t_{ign}-4$	0.6	800	10
$t_{ign}-5$	0.6	1100	10
$t_{ign}-6$	0.6	750	60
$t_{ign}-7$	0.6	1000	60
$t_{ign}-8$	1.0	800	10
$t_{ign}-9$	1.0	875	60
$t_{ign}-10$	1.0	750	60
$t_{ign}-11$	2.0	1100	10
$t_{ign}-12$	2.0	700	20
$t_{ign}-13$	2.0	800	40

2.2. Target search algorithm (TSA)

The TSA [10] is a trial-and-error method for possible further reduction of a skeletal model generated by the graph-based methods. Since TSA is a brute force method, fast computation of targeted cases are essential and these computations can be accelerated with parallel computing. The species in the starting model are ranked by their maximum concentrations under target conditions in ascending order, since low concentration intermediate species are potential candidates for removal. By testing the selected reactor model under targeted conditions, a species along with its associated reaction steps will be kept if removal of this species leads to error exceeding a threshold, typically 10–15%.

3. Issues in current automatic reduction with 0-D testbeds

As discussed above, the automatic reduction can be efficiently executed by using a 0-D testbed as the sampler in graph-based methods as well as the evaluator in trial-and-error methods. During the TSA reduction, the error evaluation is based on both the auto-ignition delay time and the adiabatic temperature. Because of the high cost of running 1-D PREMIX code especially with a large kinetic model, the laminar flame speeds are not targeted and hence species important to premixed flames may be removed. The ignition delay time is conventionally defined by 400 K increase over the initial temperature. Relative errors of 10% and 15% for the ignition delay time are respectively used in the DRGEP and TSA methods. In order to include different types of compression ignition in piston engines, thirteen target conditions [16] from the analysis of gasoline compression ignition engine are listed in Table 1. The aforementioned 6-sample extraction is applied to each case. The phrase “Dijkstra-JAC” is used to stand for the DRGEP method using Jacobian-aided pairwise relation with the Dijkstra's shortest path algorithm. The target species at the beginning of the search are fuel, oxidizers (O_2 and N_2), and products (CO_2 and H_2O).

Figures 3 and 4 compare computed laminar flame speeds among detailed and skeletal mechanisms from the automatic reduction using the DRGEP method as well as the further reduced skeletal mechanisms using the TSA with the 0-D WMR testbed. The starting detailed chemical models of methane and propane are the GRI-3.0 [29] with 53 species and that by Qin et al. [30] with 70 species respectively. Both models were developed to give reasonably good predictions of the laminar flame speeds over a wide range of equivalence ratios, typically from 0.6 to 1.6. As seen in Figs. 3 and 4, the two skeletal models (33 species for methane and 42 species for propane) by the graph-based method (DRGEP_JAC) after the first step capture the trends of the laminar flame speed versus equivalence ratio well, but there is still room for improvement. The worst-case error of DRGEP_JAC is 8%

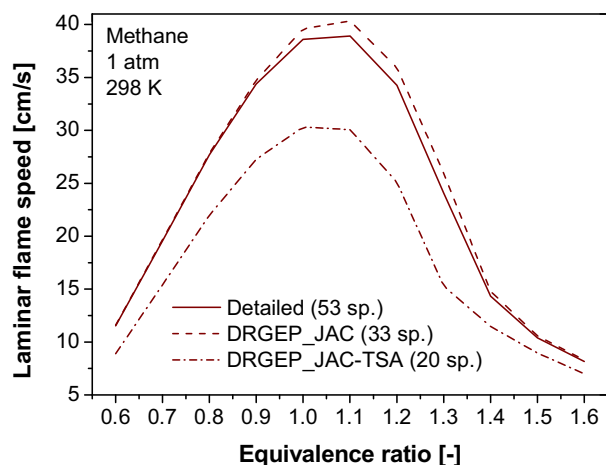


Fig. 3. Comparison of methane laminar flame speed prediction under 1 atm and 298 K among detailed model, and skeletal models from Jacobian-aided DRGEP (DRGEP_JAC) method and subsequent TSA method (DRGEP_JAC-TSA). The detailed model used for methane is GRI-3.0 [29] with 53 species.

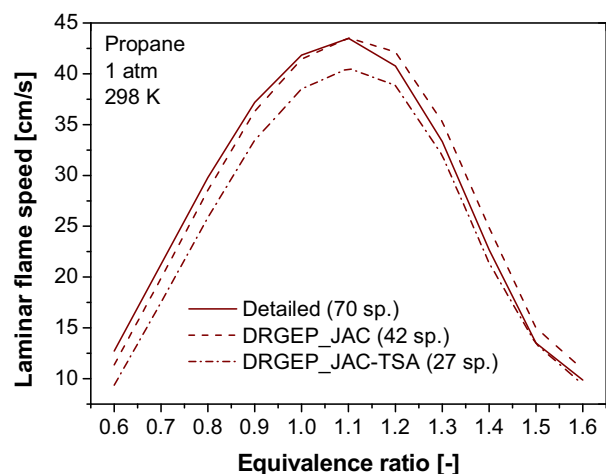


Fig. 4. Comparison of propane laminar flame speed prediction under 1 atm and 298 K among detailed model, and skeletal models from Jacobian-aided DRGEP (DRGEP_JAC) method and subsequent TSA method (DRGEP_JAC-TSA). The detailed model used for propane is by Qin et al. [30] with 70 species.

and 11% for methane and propane respectively. These two skeletal models are further reduced by the TSA (DRGEP_JAC-TSA), generating two smaller skeletal mechanisms (20 species for methane and 27 species for propane). Poor predictions of the flame speeds are seen with these two smaller skeletal models. Similarly for n-heptane as presented in Fig. 5, the graph-based skeletal reduction is capable of retaining species important to the laminar flame. However, the subsequent trial-and-error method using 0-D testbed removes some species important to flame chemistry. In Fig. 5, the Dijkstra-JAC and AStar-JAC mean the TSA-reduced models are based on the skeletal models developed from the Jacobian-aided DRGEP approach with Dijkstra and AStar graph search algorithms. In short, without including flame speeds as targets, further reduction with TSA is not capable of keeping species important to flame chemistry. The following sections aim at exploring answers to two questions: 1) How can the current DRGEP-based methods be improved for more accurate predictions of flame speeds? 2) With an accurate skeletal model developed, say by DRGEP-based approaches, how can the trial-and-error methods be used for further reduction without setting flame speed as targets?

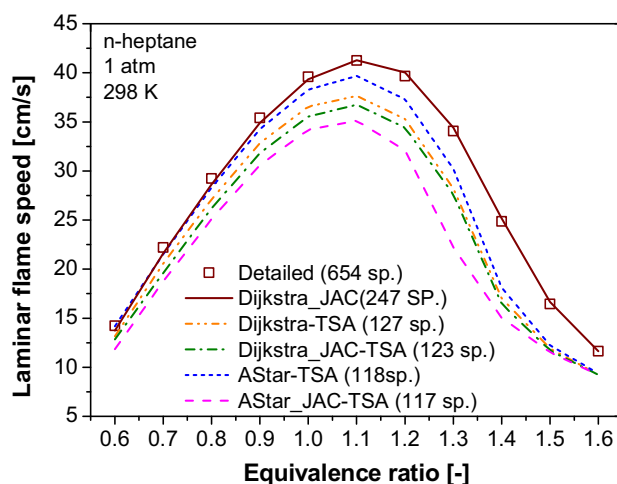


Fig. 5. Comparison of n-heptane laminar flame speed prediction under 1 atm and 298 K among detailed model, and skeletal models from Jacobian-aided DRGEP method (Dijkstra_JAC in [15]) and subsequent TSA-reduced skeletal models from various DRGEP-based methods in [15].

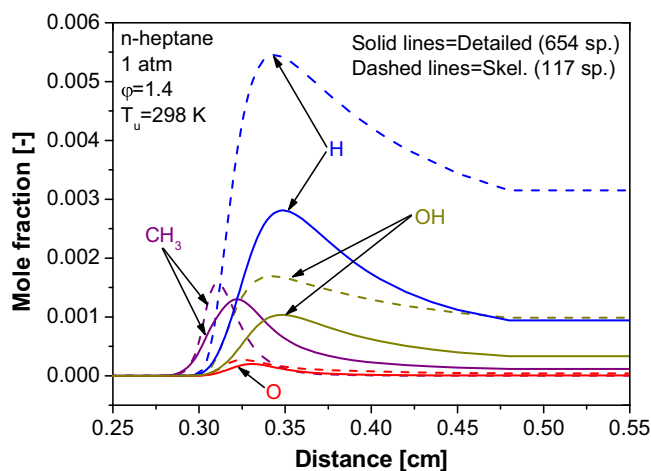


Fig. 6. Comparison of key intermediate species profiles between detailed and TSA-reduced 117-species n-heptane skeletal model of Fig. 5 in a laminar flame case under 1 atm, 298 K, and ϕ of 1.4.

4. Improvement on the automatic reduction methodology to yield accurate flame speeds

4.1. Improvement on the graph-based methods

Many earlier studies have been conducted to determine the key species controlling flame behaviors of various fuels. Kelley et al. [31] demonstrated that the structures of the laminar premixed flames from n-pentane to n-alkanes are similar since, the corresponding sensitivity analysis identifies the same controlling reactions in the kinetic models. Ji et al. [32] conducted a sensitivity analysis for n-C₁₂H₂₆ reaction coefficient and binary diffusion coefficient from lean to rich, revealing that $H + O_2 \rightarrow O + OH$ is the most important reaction, and OH, H and O are the three key radicals most affecting laminar flame speeds. Additionally, by reviewing the relevant publications, Ranzi et al. [33] concluded that for hydrocarbon and oxygenated fuels, the hydrogen radical (H) significantly enhances flame speed by its branching activation, and methyl (CH₃) is the most effective inhibitor by removing H via the combination reaction to form methane (CH₄). Fig. 6 compares the computed profiles of these key radicals under equivalence ratio of

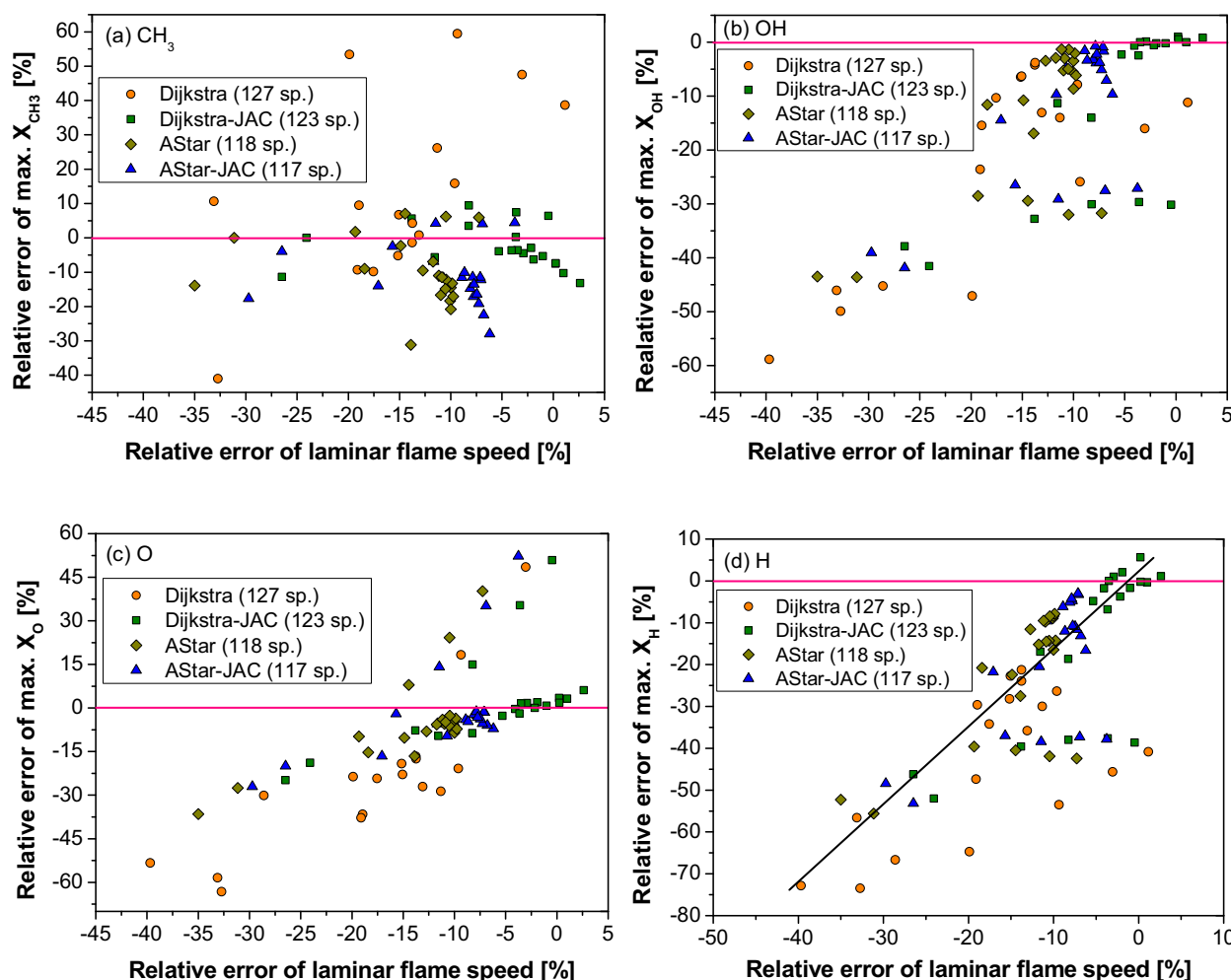


Fig. 7. Relative error of maximum temperature or radical mole fraction vs. relative error of laminar flame speed: (a) CH_3 ; (b) OH; (c) O; (d) H. The skeletal models used are from Fig. 5.

$\varphi = 1.4$ using PREMIX [34] with the detailed n-heptane model and the 117-species skeletal model in Fig. 5. As seen in Fig. 6, large discrepancies exist in the predicted peak values for these key radicals, especially for H and OH.

For better quantifying the relation between the radicals and the flame speed, we evaluate the correlation between the relative errors of key radical maximum mole fractions and the corresponding errors in laminar flame speeds. According to the suggestions by previous studies mentioned above, CH_3 , OH, O, H are selected as the candidates. Four TSA-reduced skeletal models in Fig. 5 are used to explore the correlation. PREMIX runs were conducted with φ from 0.6 to 1.6 by 0.1 interval, under two initial conditions: (1) $T = 298 \text{ K}$, $P = 1 \text{ atm}$; (2) $T = 353 \text{ K}$, $P = 10 \text{ atm}$. The results are presented as scatter plots of relative errors of species versus the relative errors of flame speeds in Fig. 7. From the plot of 7a, the relative errors of CH_3 mole fraction exhibit random patterns without obvious trends. The relative errors of maximum mole fraction of OH and O (in plots 7b and 7c) are seen to show positive correlation with the relative errors in flame speeds. However, in the flame speed region with acceptable error such as 0–15%, the relative errors of these two radicals exhibit two branches and the branch with denser data becomes uncorrelated with the flame speed error. In comparison, the relative errors of H correlate better with the relative errors of flame speeds in 7d. The above analysis suggests H radical be added as an additional search target in the graph-based methods to enhance the accuracy of flame speed predic-

tion. Several previous studies [35,36] added radicals such as HO_2 , CH_2O , OH and CO to the search targets for better keeping low-temperature ignition chemistry because in the NTC region, it is difficult to achieve accurate large-scale reduction without these additional species related pathways. Numerical explorations on various chemical kinetic models reveal that search targets with fuel and H are sufficient to keep the flame chemistry.

4.2. Using laminar speed sensitivity for identifying important species

After an accurate starting skeletal model is developed by the graph-based method, further automatic reduction considering flame speeds will be conducted. As alluded to earlier, trial-and-error methods with repeated time-consuming running of 1-D laminar flame may be impractical especially with a large starting skeletal model, such as one with more than 500 species for gasoline surrogates [16] after the DRGEP-based method. Since the normalized sensitivity of flame speeds with respect to the reaction rate constant is used routinely to identify controlling steps, this approach is explored for identifying species that are important to flame chemistry. The PREMIX [34] code is used to compute the sensitivities of mass flux, species, and temperature with respect to rate constant. These sensitivities are used to deduce the normalized sensitivity of flame speed with respect to rate constant, $\partial \ln S_L / \partial \ln r_k$, where S_L is the laminar flame speed and r_k is the rate constant of reaction step k. With the normalized flame speed

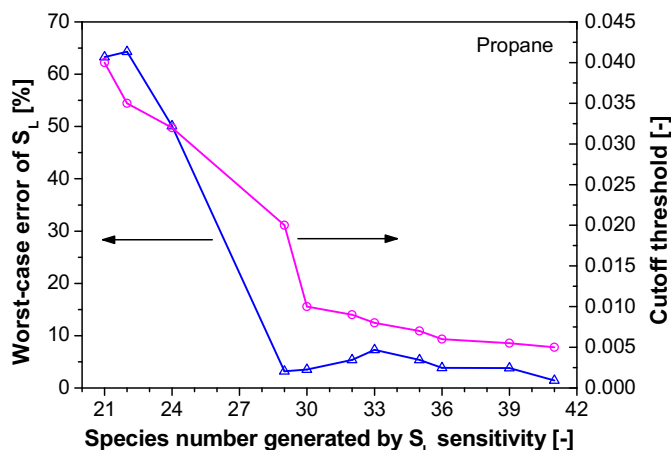


Fig. 8. Worst-case error of flame speed and cutoff threshold vs. species number generated by the flame speed sensitivity ($\partial \ln S_L / \partial \ln r_k$). Propane is the fuel with 70-species detailed model by Qin et al. [30]. Conditions of $\varphi = 0.6$ –1.6 with 0.1 interval, 1 atm, and 298 K are used for the worst-case error comparison. Sensitivity analysis is conducted under $\varphi = 0.7, 1.0$ and 1.6, 1 atm and 298 K.

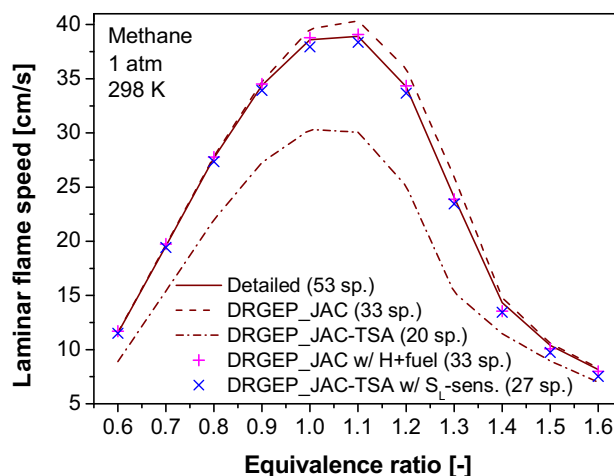


Fig. 10. Comparison of laminar flame speeds of methane between the improved and unimproved automatic reduction approaches in Fig. 3.

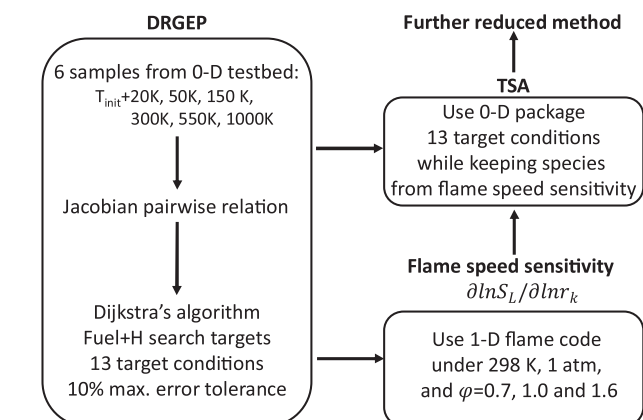


Fig. 9. An improved reduction methodology for both laminar flame and auto-ignition.

sensitivity, the relative importance of a step is gauged against a threshold. A reaction step with normalized sensitivity exceeding the threshold is considered important to the prediction of flame speeds. Therefore, with a threshold, the important steps can be identified and species in these reaction steps are also considered important to the computation of flame speed. Let's refer this set of species as flame speed set and based on this set, a skeletal model can be constructed. The size of flame speed set is expected to decrease with increasing threshold value. Therefore it is useful to explore the largest threshold value that will produce a skeletal model with acceptable accuracy.

In the exploration, flame speed sets are gathered by setting a threshold ranging from 0.001 to 0.01. Three flame cases with equivalence ratio of $\varphi = 0.7, 1.0$ and 1.6 at 1 atm and 298 K are computed using PREMIX [34]. In Fig. 8, an example of propane is given to show the worst-case relative errors in the computed flame speeds versus the number of species in the flame speed set identified by the flame speed sensitivity. Also plotted is the cut-off threshold against the species number generated by the flame speed sensitivity. As expected, less species are kept in the flame set as the threshold increases in a monotonic manner. Also revealed in Fig. 8, the worst-case error is correlated with the cutoff threshold and a skeletal mechanism with 29 species is found to predict flame speeds with less than 10% relative error in this case.

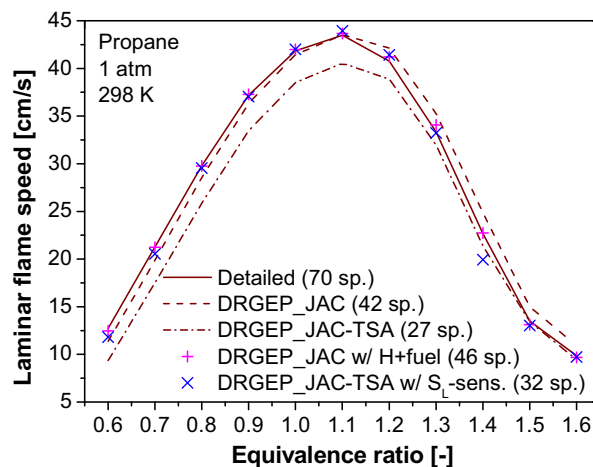


Fig. 11. Comparison of laminar flame speeds of propane between the improved and unimproved automatic reduction approaches in Fig. 4.

To develop a skeletal model for general use in both flame transport and ignition purposes, TSA with 0-D testbed will be conducted while keeping species in the flame speed set. In Fig. 9, an improved methodology in mechanism reduction is suggested.

5. Comparisons between the improved and current automatic reduction methodologies

With the improved methodology, development of starting skeletal models was carried out with additional H as the target species using the DRGEP-JAC method. The goal is to develop a starting skeletal model with improved accuracy for flame speed predictions. Next the starting skeletal model is used to determine the flame speed set. Lastly, further reduction by the TSA using the 0-D WMR testbed is conducted with the flame speed set kept. Figures 10–12 compare computed flame speeds versus equivalence ratio with the improved and current approach for methane, propane and n-heptane (Dijkstra-JAC in [15]) for the unimproved method) respectively. In these three figures, the cross symbols denote results with the improved approach using DRGEP-based methods to retain flame chemistry by including H as an additional search target. From Figs. 10–12, one can conclude that the addition of H to the target list improves the accuracy of starting skeletal model in the prediction of flame speeds over a wide range of equivalence ratios.

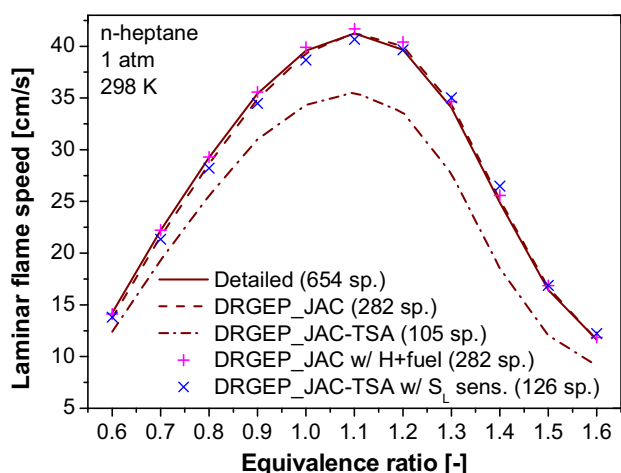


Fig. 12. Comparison of laminar flame speeds of n-heptane between the improved and unimproved automatic reduction approaches (Dijkstra-JAC in [15]) in Fig. 5.

Next, based on the improved DRGEP-based reduction, the flame speed sensitivity is computed for the three fuels to generate flame speed sets of 27-species, 29-species, and 51-species for methane, propane and n-heptane respectively.

Finally, the TSA is conducted with the 0-D WMR targeted for 15% ignition delay time worst-case error. In this round of reduc-

tion, species in the flame speed set is kept while remaining species are tested for possible removal. The symbols “x” in Figs. 10–12 correspond to flame speeds computed with the further reduced skeletal models using the improved methodology. The results confirm that the flame speed predictions are enhanced by keeping the species in the flame speed set. Simultaneously, in Fig. 13, accurate predictions of auto-ignition delays under engine conditions are ensured by the TSA for the three fuels with high fidelity as expected.

6. Application to primary reference fuel (PRF) skeletal reduction

The improved automatic reduction scheme is used for generation of skeletal models for primary reference fuel (PRF) surrogate gasoline, intended for engine applications. A 1033-species LLNL v2 detailed PRF model [37] is chosen for reduction. To develop a PRF reduced model for wide applications, PRF0 (0% iso-octane in the mixture by vol.), PRF35, PRF70, PRF85, and PRF100 are among the target fuels to cover the various degrees of blending. After the implementation of the newly improved automatic reduction with H as an additional target species, a 516-species skeletal model is first obtained by the DRGEP method. This is followed with a 92 species pool relevant to flame speed by the sensitivity analysis. Finally, a 207 species model is obtained by the TSA method by keeping the 92 flame related species. In Figs. 14 and 15, the performances of the 207-species skeletal model are assessed for PRF35 and PRF85 in terms of laminar flame speeds and auto-ignition delays over a

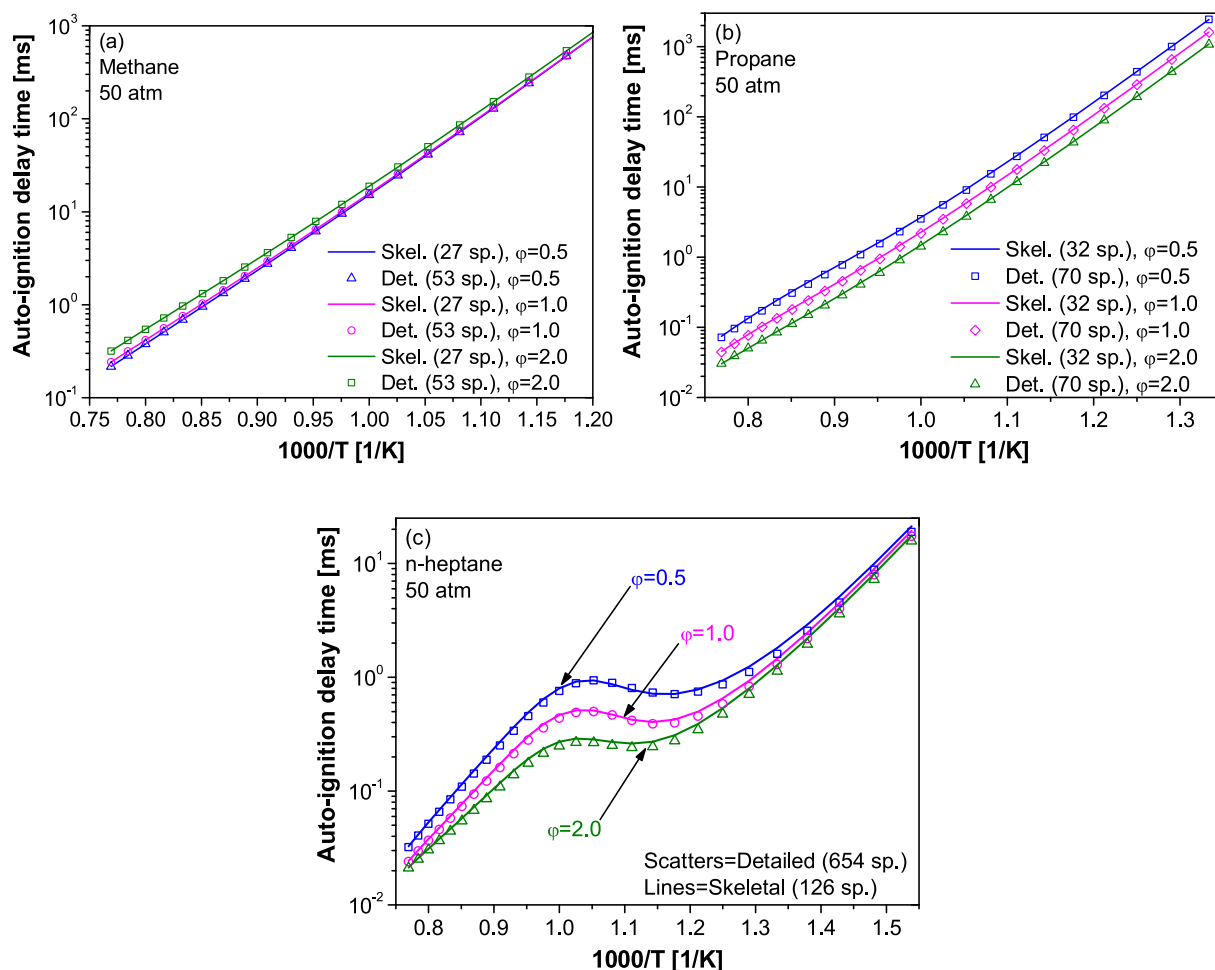


Fig. 13. Comparison of computed auto-ignition delay time among detailed models and skeletal models developed by the new scheme under 50 atm and wide ranges of temperature and equivalence ratio, (a) methane, (b) propane, (c) n-heptane.

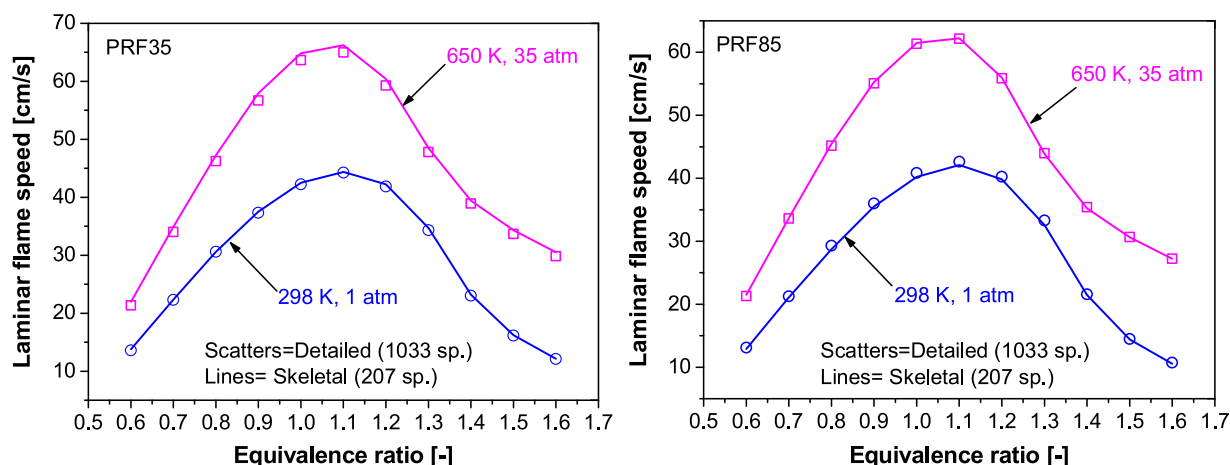


Fig. 14. Comparison of laminar flame speeds of PRF35 and PRF85 between detailed model [37] and 207-species skeletal model developed from the improved automatic reduction.

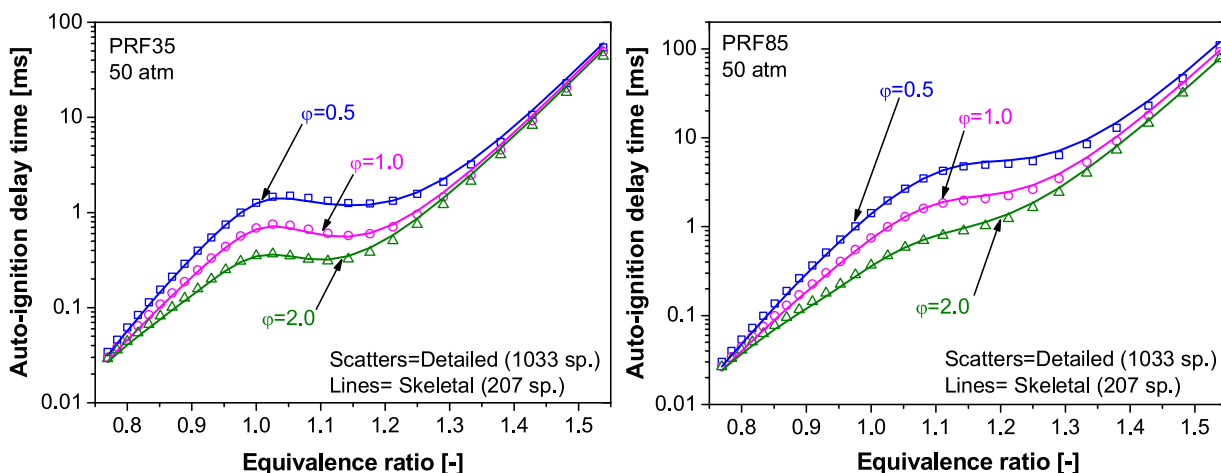


Fig. 15. Comparison of auto-ignition delay time of PRF35 and PRF85 between detailed model [37] and 207-species skeletal model developed from the improved automatic reduction.

wide range of conditions. The results in these figures confirm that robust features in the skeletal model in accurate prediction of combustion processes under engine conditions.

7. Conclusion

In this study, current automatic reduction approaches for chemical kinetic models are improved by considering both ignition delays and laminar flame speeds. The major results and findings are:

- From the analysis of the correlation between errors of key radical maximum mole fractions and laminar flame speeds, the accuracy of H radical is found most important to the accuracy in predicting laminar flame speeds for hydrocarbon fuels. With addition of H to the search target in graph-based methods (DRGEP in this study), a starting skeletal model can be constructed with improved accuracy in laminar flame speed prediction.
- By analyzing the normalized flame speed sensitivity, a method to identify a group of species important to prediction of flame speed is proposed. Numerical explorations with skeletal models generated with this set of species with different cut-off threshold were conducted to confirm the validity of this concept.

- Trial-and-error methods (TSA in this study) with a 0-D testbed can be used to further reduce the starting skeletal models with accurate predictions of both ignition delay and flame speed by keeping the species pool generated by the flame speed sensitivity during the reduction.
- The improved automatic reduction methodology is assessed for several hydrocarbon fuels of practical interest. In this study, a 27-species methane, a 32-species propane, a 126-species n-heptane, and a 207-species PRF skeletal models are developed showing prediction of both ignition delays and flame speeds in close agreement with those from the respective detailed models.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.combustflame.2017.11.024](https://doi.org/10.1016/j.combustflame.2017.11.024).

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